

Holyfaith Academy, Dhuliyān

Semester-I Examination

Subject: Chemistry

Class IX

Time: 2 Hours

Full Marks: 80

All questions are compulsory unless specified otherwise. Read the instructions carefully.

Section - 1 (40 Marks)

(Compulsory - Attempt all questions)

Question 1

A. Multiple Choice Questions (MCQs): ($15 \times 1 = 15$ marks)

1. Which of the following is an example of a physical change?
 - a) Rusting of iron
 - b) Melting of wax
 - c) Burning of magnesium ribbon
 - d) Digestion of food
2. Which of the following is a covalent compound?
 - a) NaCl
 - b) MgO
 - c) CH₄
 - d) CaF₂
3. The formula of quicklime is:
 - a) Ca(OH)₂
 - b) CaO
 - c) CaCO₃
 - d) CaCl₂
4. Which gas is liberated when zinc reacts with dilute sulphuric acid?
 - a) Oxygen
 - b) Carbon dioxide

c) Hydrogen

d) Nitrogen

5. The atomic number of an element represents:

a) Number of neutrons

b) Number of protons

c) Mass number

d) Number of electrons + protons

B. Fill in the blanks: (1 mark each)

6. The process of converting water vapour directly into ice is called _____.

7. The valency of Aluminium in Aluminium oxide _____.

8. The two radicals of calcium sulphate are _____ and _____.

C. True or False: (1 mark each)

9. Electrovalent bonds are formed by the sharing of electrons. (True/False)

10. Decomposition of silver nitrate on heating is a chemical change. (True/False)

D. Name the following: (1 mark each)

11. The property of water is to dissolve many substances.

12. The ratio of hydrogen to oxygen atoms in water.

13. The particle in an atom with a negative charge.

E. One-word answer: (1 mark each)

14. Name a compound used in the laboratory for drying gases.

15. The smallest particle of an element which takes part in a chemical reaction.

Section - 2 (40 Marks)

(Answer any four questions. Each question carries 10 marks.)

Question 2

A. Write the balanced chemical equations for the following: (5 marks)

i) $\text{Mg} + \text{HCl} \rightarrow ?$

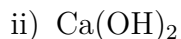
ii) Decomposition of potassium chlorate in presence of MnO_2 .

iii) ZnCO_3 heated strongly

- B. 10 ml of alcohol is added to water to make a solution having a total volume of 250 ml. Calculate the percentage of alcohol in the solution (by volume). (2 marks)

Question 3

- A. Calculate the molecular mass of: (5 marks)



- B. Define: (2 marks)

i) Hygroscopic substance

ii) Isotopes

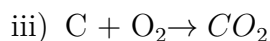
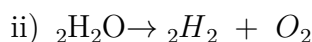
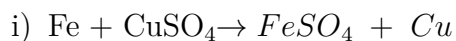
- C. Name one example each of: (1 mark)

i) A precipitation reaction

ii) A double displacement reaction

Question 4

- A. Identify the type of reaction: (5 marks)



- B. Arrange in the increasing order of reactivity of these metals: (3 marks) Na, Zn, Al, Cu, Mg, Fe

Question 5

- A. Give the reason: (5 marks)

i) Why is sodium metal kept in the kerosene container? (2 marks)

ii) Why does ice float on the surface of the water? (2 marks)

- B. State three ways in which rusting of iron can be prevented. (3 marks)

Question 6

- A. Write the electronic configurations of: (5 marks)

i) Sodium atom

ii) Oxygen atom

- B. Differentiate between hard water and soft water. (3 marks)

Question 7

- A. Explain the following terms with examples: (4 marks)
- i) Specific heat capacity of water
 - ii) Dehydrating agent
 - iii) Catalyst
 - iv) Efflorescence
- B. List four properties of covalent compounds. (4 marks)
- C. State two advantages of water as a solvent in biological systems. (2 marks)

Question 8

- A. Write the electrical configuration of potassium. Draw the structure and mention where electrons, protons, and neutrons are located in this atom. (5 marks)
- B. Differentiate between: (3 marks)
- i) Electrovalent bond and covalent bond (two points)
 - ii) Thermal decomposition and electrolytic decomposition (two points)
- C. Write an example of each: (2 marks)
- i) A drying agent
 - ii) An anhydrous salt

Question 9

- A. Explain: Which observation in the discharge tube experiment indicates that: (5 marks)
- 1. Cathode rays consist of negative charge particles
 - 2. Cathode particles possess momentum
- B. Define: (3 marks)
- i) Exothermic reaction
 - ii) Displacement reaction
 - iii) Photochemical reaction
- C. Write the balanced chemical equation for: (2 marks)
- i) Action of cold water on sodium
 - ii) Combustion of methane

Question 10

- A. Explain with examples: (5 marks)
- i) Neutralisation reaction
 - ii) Oxidation reaction
 - iii) Combination reaction
- B. What is meant by the 'water of crystallisation'? Give two examples of substances containing it. (3 marks)
- C. State two disadvantages of hard water. (2 marks)